Reading the IUT dispute through indexed identifications: an educational model in undergraduate mathematics

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Abstract

The dispute between Scholze–Stix and Mochizuki over Corollary 3.12 of inter-universal Teichmüller theory (IUT) III has been described in words: are isomorphic copies to be identified or not? This note gives an educational model that states the *shape* of the dispute precisely using only undergraduate mathematics—sets and maps, multiplication of positive real numbers, graphs, and inequalities. Instead of the equality sign we use a symbol $x \approx_f y$ that records which identification is used, and we declare four rules at the outset: identity and inverse, composition, action on numbers, and sets of indices. In this notation an identification that does not return to 1 around a loop (a monodromy) becomes a formula. Both Mochizuki's own elementary numerical model and the three options described by Scholze–Stix (omit j^2 and the inequality is empty; keep it and the identifications are inconsistent; blur, and the blurring must be by a factor of at least $O(l^2)$) then sit on one line, the inequality $(k-1)Q \leq E$. What is agreed is that an identification which does not return to 1 around a loop leads to a contradiction if it is used consistently. What is disputed is whether the "license", in Mochizuki's word, to identify up to "relatively small error terms" can be obtained independently of the conclusion. The model does not decide this question.

Part II, new in this version, opens the single number E. Read backwards from the public Lean formalization lana-agents/iut, the notation needs four more things: values on the components of tensor packets; indices that pair a permutation of the factors with identifications on the components; a rule for the order in which the hull of a union and an average are taken; and a record of the construction that a compared number comes from. In a model with one prime, its places and the label permutations of (Ind1), the hull over all positions of the theta value is governed, at each component, by the smallest q-order among the factors, whereas the computation of IUT IV uses an average. After averaging over the components the difference is an expected order statistic, and it vanishes exactly when all places above the prime have the same q-order. For every reading of the union, Corollary 3.12 together with the upper bound of IV implies the height bound: in one reading the content is carried by the upper bound of IV, in the other by Corollary 3.12 itself. Which reading the source intends is not decided. A companion paper states the model of Part I in homotopy type theory; the quantitative core of Part II is checked in Lean.

1 Introduction

1.1 What the dispute is about

Mochizuki's inter-universal Teichmüller theory (IUT) is a theory that claims to imply the abc conjecture. One of its central steps is Corollary 3.12 of IUT III, an inequality that compares two numbers through the Θ-link, which connects two "stages" (Hodge theaters). Scholze and

Stix objected to this comparison [SS]. The point in question is whether isomorphic objects appearing on the two stages are to be identified or not. Scholze and Stix write:

> S "As we are interested in comparing real numbers, and we have seen that various copies of ordered 1-dimensional $\mathbb{R}$-vector spaces arise, it is critical to spell out all identifications of copies of real numbers that are in place. In particular, in order to say that the abstract Θ-pilot object encodes the arithmetic degree of the (j-th) concrete Θ-pilot object, we saw that it was necessary to change the isomorphism $\mathbb{R} \cong \mathbb{R}_{\odot,\Theta}$ by the scalar j^2, $j = 1, \ldots, \ell^*$ (or their average, if one is interested in the averaged degree)."
>
> [SS, §2.2]

Mochizuki, for his part, uses different symbols for different kinds of equality: the gluing by the Θ-link is written "=", the "license" to identify up to relatively mild indeterminacies is written $\overset{!!}{\leftrightarrows}$, and the approximate relation one wants to obtain from it is written "$\approx$".

> S "the 'q^N' in *one copy of scheme theory* is **identified** with the 'q' in the *other copy of scheme theory*. This gluing is referred to as the **Θ-link**. Thus, the 'q^N' on the *left-hand side* of the Θ-link is *glued* to the 'q' on the *right-hand side* of the Θ-link, i.e.,
>
> $$q^N_{\mathrm{LHS}} \quad \text{`}=\text{'} \quad q_{\mathrm{RHS}}$$
>
> [cf. §3.3, (vii), for more details]. Here, 'N' is in fact taken *not* to be a *fixed natural number*, but rather a sort of **symmetrized average** over the values j^2, where $j = 1, \ldots, l^*$" ...
>
> "up to **relatively mild indeterminacies** — i.e., *'relatively small error terms'* — $\{q^{j^2}_{\mathrm{LHS}}\}_j$ may be **'confused'**, or **'identified'**, with $\{q^{j^2}_{\mathrm{RHS}}\}_j$, that is to say,
>
> $$\{q^{j^2}_{\mathrm{LHS}}\}_j \quad \overset{!!}{\leftrightarrows} \quad \{q^{j^2}_{\mathrm{RHS}}\}_j$$
>
> ... Once one is equipped with this **'license'** to *confuse/identify* $\{q^{j^2}_{\mathrm{LHS}}\}_j$ with $\{q^{j^2}_{\mathrm{RHS}}\}_j$, the derivation of the desired *approximate relation*
>
> $$\{q^{j^2}\}_j \quad \text{`}\approx\text{'} \quad q$$
>
> and hence of the desired **bounds** on **heights** is an *essentially formal matter*"
>
> [Alien, Introduction]

1.2 The plan of this note

This note writes these "identifications" with a symbol that carries, as an index, the identification that is used. Only undergraduate mathematics is needed: sets and maps, the group of positive real numbers under multiplication, graphs (vertices, edges, cycles) and inequalities; Part II also uses permutations of a finite set and weighted averages. No category theory, homotopy type theory or anabelian geometry is used.

Part I gives each container a single number.

- Section 2 looks at identifications that do not return around a loop, using unit conversion and the angle on a circle.
- Section 3 defines indexed identifications and their four rules.
- Section 4 asks, on Mochizuki's three graphs (segment, loop, universal cover), when consistent values exist.
- Section 5 sets up two paths, the Θ-link and a natural identification, and reduces the three options of Scholze–Stix to a one-line inequality.
- Section 6 rereads Mochizuki's elementary numerical model in the same notation.

- Section 7 shows how the shape of a height bound comes out of Corollary 3.12 together with the upper bound of IUT IV.
- Section 8 puts the two positions in one table, and Section 9 compares, one level down, the presented reals of [U].

Part II, new in version 2.0.0, opens the error E.

- Section 10 explains why one number is not enough, and Section 11 extends the notation.
- Section 12 sets up a model with components, and Section 13 computes the hull in two readings of the union.
- Section 14 writes the dispute in the extended notation, and Section 15 gives the correspondences and states what remains.

Section 16 summarizes, Section 17 gives exercises, and Appendix A lists the Lean declarations that check Part II.

1.3 What this note does not claim

- It makes no judgement on Corollary 3.12, on Theorem 1.10 of IUT IV, or on the abc conjecture, and none on which position is right.
- The model follows the simplification of Scholze–Stix: the containers in which numbers are measured are regarded as ordered one-dimensional real vector spaces [SS, §2.1.4]. Mochizuki, on the other hand, counts it among the "*erroneous attempts by SS to simplify IUTch* ... to such an extent that it leads to meaningless contradictions" [R18, §4]. The model is a tool for seeing the shape of the dispute; it is not IUT itself.
- The content of the multiradial algorithm (Theorem 3.11 of IUT III) and of anabelian geometry is not treated. In Part I the size of the error they produce is represented by a single number E. In Part II the part of the error produced by the label permutations of (Ind1) is computed in a model, and the part produced by (Ind2) and (Ind3) is still a single number.
- The model of Part II has one prime, the places above it and the label permutations; regions are represented by their log-volumes. The regions themselves, the differents, the archimedean places and the normalizations of IV are not modelled.

1.4 Relation to earlier explanations

Elementary explanations of the dispute have been given by the parties themselves. Mochizuki gave the analogy of the coordinate charts of a manifold [R18, (LbEx5)], the example of the logarithm in high-school mathematics [Alien, §3.11 (ii)], and an elementary numerical model with real numbers A, B, ϵ [Alien, Example 3.11.4]. Scholze and Stix stated their objection with copies of ordered one-dimensional real vector spaces and a diagram of their identifications [SS, §2.2]. Nothing in the mathematics of this note is new; every proposition is at the level of an undergraduate exercise. What the note adds is to declare the rules of identification first and to put the explanations of both sides into the same notation.

The companion paper [Ur] states the same model in homotopy type theory, checks its quantitative core in Lean, and discusses the relation with presented reals and Booij's locators. This note is its educational entry point; many of its propositions are translations of propositions of [Ur] into undergraduate language, and the correspondence is recorded at each place. The correspondence between the diagrammatic language of Project LANA (the commutativity of diagrams and the compatibility (9-1)), the notation of this note and the identity types of homotopy type theory is given in [Ur, §6.4].

Part II follows the public Lean formalization [LANA-iut], which states two readings of the union in Corollary 3.12—the theta value at any label of $\mathbb{S}^{\pm}_{j+1}$, or at the label j as in IUT IV—and records that under the first the hull is governed by a minimum rather than by an average. That the actual (Ind1) includes permutations between tensor factors, which a simplified computation may omit, was noted earlier by Dupuy–Hilado [DH, §6.2, footnote 7]. An explicit initial Θ-datum for which, under the readings stated there, the union in the literal reading is not contained in the module used in Step (v) is given in [Note]. In the setting of Part I, the inequality of Proposition 7.1 is stated in [TK] as the condition for the containment used in Step (v) of IUT IV. What Part II adds is the undergraduate formulation, the size of the difference between the two readings, and Theorem 14.3.

1.5 Labels

The labels in the text mean the following.

S A statement of the source (a quotation).

P A proposition proved in this note. A proposition with P only is either an elementary fact or a known fact stated in its standard form (in the latter case its source is given).

R Not new mathematically, but the notation, formulation or correspondence is that of this note. For a proposition it is used together with P, and the source of the underlying mathematics is given as "Known mathematics".

N A theorem first proved here (as far as the author has checked). There is no proposition labelled N in this note.

H A reading or interpretation of this note (not a proof).

Definitions and examples without a label are standard or expository.

Part I
One number

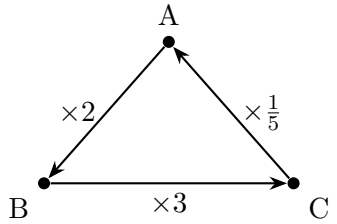
2 Identifications that do not return around a loop

2.1 Unit conversion

Example 2.1 (A conversion table)**.** A quantity is measured in three units A, B and C. Suppose the conversion table is

$$x_B = 2\,x_A, \qquad x_C = 3\,x_B, \qquad x_A = \tfrac{1}{5}\,x_C,$$

where x_A is the value measured in A, and so on. Going from A through B and C back to A multiplies a value by $2 \cdot 3 \cdot \frac{1}{5} = \frac{6}{5}$.



Around the loop the factor is 6/5.

If we look for measurements x_A, x_B, x_C consistent with this table, the three equations give $x_A = \frac{6}{5}x_A$, so $x_A = 0$. *The only consistent measurement is* 0. If the last entry is changed to $\frac{1}{6}$, the product around the loop becomes 1, and every x_A extends to exactly one consistent measurement.

There is a second point. To convert a value in A into a value in C, going through B multiplies by 6, while reversing the conversion "from C to A" multiplies by 5. *Unless we say which route was used, the answer is not determined.*

2.2 The angle on a circle

Mochizuki explains, with the analogy of manifolds, that omitting labels produces apparent contradictions.

> S "(This example may, roughly speaking, be thought of as a sort of generalization of the example discussed in (LbEx1).) Consider the theory of (topological or differential) **manifolds** of dimension n, for n a positive integer. Manifolds (of dimension n) are, by definition, collections of *distinct copies* — i.e., copies equipped with **distinct labels** — of (n-dimensional) *Euclidean space* glued together by means of various *gluing isomorphisms*. In particular, it is quite possible that the use of **numerous distinct copies of Euclidean space** in this theory might impress a newcomer to this theory (such as a student) as being entirely **superfluous**. Such a newcomer might feel motivated to attempt to achieve a **'dramatic simplification'** of this theory by **identifying** these copies of Euclidean space with one another and hence to assert that the geometry of *arbitrary* (topological or differential) *manifolds* (of dimension n) may be **'reduced'** to the geometry of (n-dimensional) *Euclidean space*." ...
>
> "If one **identifies** these **distinct copies of Euclidean space** that appear in the theory of (topological or differential) manifolds with one another, then it is easy to obtain a *'contradiction'*. Such a 'contradiction' may, of course, be *misinterpreted* as an **'internal contradiction'** in the theory of (topological or differential) manifolds. In fact, however, there is **no 'internal contradiction'** in the theory of (topological or differential) manifolds; the apparent 'internal contradiction' is nothing more than a superficial consequence of the **erroneous operation** of *omitting the labels* used to distinguish the distinct copies of Euclidean space."
>
> [R18, §14, (LbEx5)]

Example 2.2 (Two charts of the circle)**.** Cover the circle by two coordinate charts. Let U_1 be the circle minus the point $(1, 0)$, with the angle $\theta_1 \in (0, 2\pi)$ as coordinate, and U_2 the circle minus $(-1, 0)$, with the angle $\theta_2 \in (-\pi, \pi)$. The overlap has two parts, the upper and the lower half; on the upper half $\theta_2 = \theta_1$, and on the lower half $\theta_2 = \theta_1 - 2\pi$.

The two coordinates are two "copies of the real line". The two parts of the overlap give two different identifications, the identity and the translation by -2π. Going once around the circle shifts the angle by 2π, so there is no continuous angle function on the whole circle that is consistent with both charts. If the two copies are not distinguished and are regarded as one real line, the apparent contradiction "$\theta = \theta + 2\pi$" appears. The circle is the simplest instance of Mochizuki's analogy.

In Example 2.1 a value is *multiplied* by a factor around the loop; in Example 2.2 it is *shifted* by an amount. From now on we treat the first form (scaling), which is the one that appears in the IUT dispute.

3 Indexed identifications

3.1 Containers and identifications

Definition 3.1 (Containers)**.** A copy of the real line carrying a label α is written $\mathbb{R}_\alpha$ and called a *container*. An element of $\mathbb{R}_\alpha$ is thought of as "a value measured in the units of α". Containers with different labels are treated as different objects, even though as sets they are the same $\mathbb{R}$.

> S A global realified Frobenioid "simply amounts to a collection of ordered 1-dimensional $\mathbb{R}$-vector spaces $\mathbb{R}_v$ parametrized by the places v of k, together with a subspace" D_0 of their direct sum "of codimension 1", subject to a normalization by the local degrees; and "These considerations show that the category of global realified Frobenioids is equivalent to the category of ordered 1-dimensional $\mathbb{R}$-vector spaces."
>
> [SS, §2.1.4]

H An ordered one-dimensional real vector space can be identified with $\mathbb{R}$ once a unit (a positive basis vector) is chosen; changing the unit multiplies values by a positive real number. In this note we use from the start containers with a chosen unit, that is, labelled copies of $\mathbb{R}$.

Definition 3.2 (Identifications)**.** R

Known mathematics: [Ur, Prop. 2.4 (ii)], [SS, §2.1.4]

An *identification* from the container $\mathbb{R}_\alpha$ to $\mathbb{R}_\beta$ is a map $f(x) = \lambda x$ given by multiplication by a positive number λ; we call $\lambda = \lambda_f$ the *scale* of f. For $x \in \mathbb{R}_\alpha$ and $y \in \mathbb{R}_\beta$ with $y = f(x)$ we write

$$x \approx_f y$$

and read it as "x and y are identified by f".

3.2 Four rules

Proposition 3.3 (Rules)**.** P R

Known mathematics: [Ur, Prop. 2.4]; for (R4), [SS, footnote 8] and [IUT3, Cor. 3.12]

(R1) Identity and inverse. $x \approx_{\mathrm{id}} x$. *If* $x \approx_f y$, *then* $y \approx_{f^{-1}} x$, *and* $\lambda_{f^{-1}} = 1/\lambda_f$.

(R2) Composition. *If* $x \approx_f y$ *and* $y \approx_g z$, *then* $x \approx_{g\circ f} z$, *and* $\lambda_{g\circ f} = \lambda_g \lambda_f$.

(R3) Action on numbers. *If* $x \approx_f y$, *then* $y = \lambda_f x$; *in particular the value itself changes under an identification. Only quantities that are unchanged by all identifications (*invariants*) may be transported without naming the identification.*

(R4) Sets of indices. *For a set* S *of identifications we write* $x \approx_S y$ *for "*$x \approx_f y$ *for some* $f \in S$*", and call*

$$S(x) := \{\, f(x) \mid f \in S \,\}$$

the set of possible images *of* x. *The composite of two such sets is* $T \circ S := \{\, g \circ f \mid f \in S,\ g \in T \,\}$.

Proof. (R1) and (R2) follow from $f^{-1}(y) = y/\lambda_f$ and $g(f(x)) = \lambda_g \lambda_f x$. (R3) and (R4) follow directly from the definitions. □

Definition 3.4 (Comparison with an error)**.** R

Known mathematics: [SS, §2.2] (blurring)

For $E \geq 0$ and real numbers a, b we write $a \leq_E b$ for $a \leq b + E$.

Remark 3.5. H The ordinary equality sign corresponds to the case in which (R1) and (R2) hold and there is always only one identification. The indexed symbol keeps in the formula which identification was used. If any of the four rules is dropped, the index becomes decoration; for example, if (R3) is ignored and values are transported without the index, we are in the situation of Proposition 3.6.

3.3 What remains when the index is dropped

Proposition 3.6. P *(elementary)*

Write $x \approx y$, dropping the index, for "$x \approx_f y$ for some identification f". Then $x \approx y$ holds if and only if x and y have the same sign (both positive, both 0, or both negative).

Proof. Since scales are positive, identifications preserve signs. Conversely, if both are positive or both negative, take $\lambda = y/x > 0$; then $f(x) = y$. If both are 0, any f will do. □

Proposition 3.7 (Invariants)**.** P *(elementary)*

For two values x, x' in one container, with $x \neq 0$, the ratio x'/x and the order relation $x \leq x'$ do not change under any identification. The values themselves, their absolute values and their differences are multiplied by λ_f.

Proof. $f(x')/f(x) = \lambda_f x'/(\lambda_f x) = x'/x$; since $\lambda_f > 0$, the order is preserved; and $f(x') - f(x) = \lambda_f(x' - x)$. □

Remark 3.8. H Comparisons inside one container do not depend on units. To compare values in different containers as numbers, one must name an identification (by Proposition 3.6, without a name only the sign survives). In the usual way of writing, this naming is hidden inside "comparing them as real numbers". Mochizuki makes the same point with the logarithm.

> S "Thus, the substantial discomfort that some high-school students encounter in accepting the notion of a logarithm may be understood as a difficulty in accepting the **identification** of the underlying additive monoid of the topological field $\mathbb{R}$ that contains the multiplicative monoid $\mathbb{R}_{>0}$ in the **domain** of log with the underlying additive monoid of the topological field $\mathbb{R}$ that appears in the **codomain** of log on the grounds that the map log is **not compatible** with [i.e., does not arise from a ring homomorphism between] the **ring structures** in its domain and codomain. Here, we recall that this identification is typically 'taken for granted' or 'regarded as not requiring any justification' in discussions concerning the natural logarithm on positive real numbers."
>
> [Alien, §3.11 (ii)]

4 Segment, loop and universal cover

4.1 Diagrams and consistent values

Definition 4.1 (Diagrams)**.** R

Known mathematics: [R18, (LbEx1)], [Ur]

A *diagram* is a finite graph with a container $\mathbb{R}_v$ at each vertex v and an identification $f_e : \mathbb{R}_v \to \mathbb{R}_w$ on each edge e from v to w. A family of values $(x_v)_v$ is *consistent* if $x_v \approx_{f_e} x_w$ for every edge $e : v \to w$. For a closed path γ, which traverses edges forwards or backwards and returns to its starting vertex, the product of the scales of its edges (the reciprocal for an edge traversed backwards) is called the *monodromy* of γ and written $\lambda(\gamma)$.

Proposition 4.2. P R

Known mathematics: [MO] (coverings are classified by the fundamental group), [R18, (LbEx1)]

Suppose the graph of a diagram is connected.

(i) *If the graph has no cycles (it is a tree), then for every vertex v_0 and every $x_0 \in \mathbb{R}_{v_0}$ there is exactly one consistent family of values with $x_{v_0} = x_0$.*

(ii) *If some closed path γ through v_0 has $\lambda(\gamma) \neq 1$, then every consistent family of values is 0 at every vertex.*

(iii) *A consistent family of values that is not identically 0 exists if and only if every closed path has monodromy 1.*

Proof. (i) In a tree there is exactly one path from v_0 to each vertex; put at each vertex the transport of x_0 along that path, and there is no other choice. (ii) Transporting x_{v_0} once around γ gives $\lambda(\gamma)x_{v_0}$, which must equal x_{v_0}; hence $(\lambda(\gamma)-1)x_{v_0} = 0$ and $x_{v_0} = 0$. Since the graph is connected, the value at every other vertex is the transport of 0, that is 0. (iii) If a family is not identically 0, then by (ii) every closed path has monodromy 1. Conversely, if every closed path has monodromy 1, choose a spanning tree and put values by (i) starting from some $x_0 \neq 0$; for an edge $e : v \to w$ outside the tree, the closed path formed by e and the tree path has monodromy 1, so $x_w = f_e(x_v)$. □

Example 2.1 is a triangular diagram whose monodromy around the loop is 6/5; by Proposition 4.2 (ii) the only consistent value is 0.

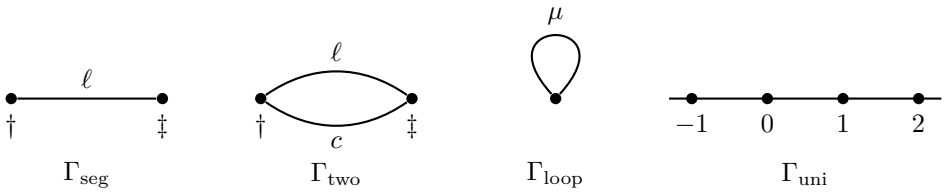
Companion paper: [Ur, Props. 3.4 and 3.8]

4.2 Mochizuki's three graphs

S "One considers the *('line segment') graph* Γ_{seg} … given by *two vertices* joined via a *single edge*. Write Γ_{loop} for the *('loop') graph* obtained from Γ_{seg} by identifying the *two distinct vertices* of Γ_{seg} — i.e., which may be thought of as *distinct labels* — to a *single vertex*. … situations where some sort of **'parallel transport'** or **'analytic continuation'** along the **loop** of Γ_{loop} gives rise to some sort of **nontrivial monodromy operator**. This sort of nontrivial monodromy may be **resolved** … *by working over* Γ_{seg} *as opposed to* Γ_{loop}. When, moreover, one wishes to distinguish situations that arise *prior to* and *subsequent to* multiple applications of the monodromy operator, it is natural to consider the *universal covering* Γ_{uni} … of Γ_{loop}" …

"one may understand the **incompatibility** of the **Θ-link** … with **arithmetic degrees** … as a sort of **nontrivial monodromy** around the **loop** … that arises as soon as one *identifies* the *domain* and *codomain* of the Θ-link — an incompatibility … that is in fact **avoided** in IUTch … precisely by **distinguishing** *the (ring structures in the) domain and codomain of the Θ-link*, i.e., by working with 'Γ_{seg}', and then applying the **multi-radial algorithms** of IUTch to obtain an alternative way to **compute this nontrivial monodromy**"

[R18, §14, (LbEx1)]



H In terms of diagrams this reads as follows.

- *Segment* Γ_{seg}. The vertices †, ‡ are joined by one edge ℓ. There is no cycle, so values are unconstrained (Proposition 4.2 (i)).
- *Two paths* Γ_{two}. The same two vertices are joined by two edges ℓ and c. The closed path that goes from † to ‡ along ℓ and back along c reversed has monodromy $\lambda_c^{-1}\lambda_\ell$.

- *Loop* Γ_{loop}. Merging † and ‡ into one vertex along the edge c turns the remaining edge ℓ into a loop, with scale $\lambda_\mu = \lambda_c^{-1}\lambda_\ell$ of $\mu := c^{-1} \circ \ell$. If this is not 1, the only value is 0 (Proposition 4.2 (ii)).
- *Universal cover* Γ_{uni}. Label the vertices by integers n and put μ on each edge from n to $n+1$. There is no cycle, so every x_0 determines exactly one consistent family $x_n = \lambda_\mu^n x_0$ (Proposition 4.2 (i) and Exercise 4). The label n remembers how many times one has gone around.

Companion paper: [Ur, Props. 3.4–3.9]

4.3 The loop with the labels dropped

Proposition 4.3. P R

Known mathematics: [IUT3, Remark 3.11.1 (vii), (viii)]

Let $0 < m < 1$ *and* $x_0 < 0$, *and put*

$$O := \{\, m^n x_0 \mid n \in \mathbb{Z} \,\}.$$

This is the family of values on the universal cover with the labels n *dropped, that is, the set of possible images of* x_0 *under* $\{\mu^n \mid n \in \mathbb{Z}\}$ *(with* $\lambda_\mu = m$*). Then* $\sup O = 0$ *(there is no maximum) and* $\inf O = -\infty$. *Hence the smallest interval containing* O *is* $(-\infty, 0)$, *and for a negative number* a *the statement "*$a \le \sup O$*" always holds and says nothing.*

Proof. Every element is negative; $m^n x_0 \to 0$ as $n \to +\infty$ and $m^n x_0 \to -\infty$ as $n \to -\infty$. □

S "one is led [at least in the absence of more sophisticated machinery!] to regard '$\underline{\underline{q}}$' as being **only well-defined up to possible confusion with '$\underline{\underline{q}}^{\lambda^n}$'**, for some indeterminate $n \in \mathbb{Z}$. That is to say, in summary, this approach gives rise to a sort of **'uninteresting/trivial multiradial representation** of '$\underline{\underline{q}}^{\lambda}$' via"

"$\{\underline{\underline{q}}^{\lambda^n}\}_{n\in\mathbb{Z}}$"

...

"this 'trivial multiradial representation' is **useless** from the point of view of applications such as the log-volume estimates given in Corollary 3.12 below ... for the following reasons: This **'trivial multiradial representation $\{\underline{\underline{q}}^{\lambda^n}\}_{n\in\mathbb{Z}}$'** is obtained by · allowing for **indeterminacies** in the **value group** portion [i.e., '$\underline{\underline{q}}^{\mathbb{Z}}$'] of the data under consideration, ..."

"By contrast, the *log-volume estimates* of Corollary 3.12 below rely, in an essential way, on the fact that in the **multiradial construction algorithm** of Theorem 3.11: · the **value group** portions of the data under consideration ... are held **rigid** [i.e., are not subject to indeterminacies], · while the **unit group** portions of the data under consideration ... are subject to the **indeterminacies** (Ind1), (Ind2), (Ind3); ..."

[IUT3, Remark 3.11.1 (vii), (viii)]

H The log-volume of $\underline{\underline{q}}^{\lambda^n}$ is $\lambda^n \log \underline{\underline{q}}$, so the source's $\{\underline{\underline{q}}^{\lambda^n}\}_{n\in\mathbb{Z}}$ corresponds to the orbit of Proposition 4.3. If the labels are dropped and one may go around the loop any number of times, the comparison becomes empty.

Companion paper: [Ur, Thm. 5.5]

5 The Θ-link and two paths

5.1 The model

Setup 5.1 (The model). R

Known mathematics: [SS, §2.1.4, §2.2], [LANA, §7.1, §10.2]

Consider the following data.

(a) Containers $\mathbb{R}_\dagger$ and $\mathbb{R}_\ddagger$ for the values of log-volumes on the domain (Θ side) and on the codomain (q side) of the Θ-link.

(b) Two identifications $\ell, c : \mathbb{R}_\dagger \to \mathbb{R}_\ddagger$: ℓ is the Θ-link (it sends the Θ-pilot to the q-pilot), and c is the natural identification that comes from both sides arising from the same curve.

(c) $b \in \mathbb{R}_\dagger$ is the log-volume of the Θ-pilot and $a \in \mathbb{R}_\ddagger$ that of the q-pilot. The Θ-link gives $b \approx_\ell a$.

(d) Units are chosen so that the scale of c is 1. Put $Q := |\log q| > 0$, $a = -Q$ and $b = -kQ$, where k is the mean of j^2 over $j = 1, \dots, l^*$ with $l^* = (l-1)/2$:

$$k := \frac{1}{l^*} \sum_{j=1}^{l^*} j^2 = \frac{l(l+1)}{12}$$

(Exercise 3). Then $\lambda_\ell = 1/k$ and $\lambda_c = 1$.

(e) An error $E \geq 0$: a single number standing for the error produced by the indeterminacies of the multiradial algorithm.

Remark 5.2. H The identification c in (b) corresponds to the identification that Scholze–Stix make by "using the natural isomorphisms $\mathbb{R}_{\odot,\Theta} \cong \mathbb{R}_{\odot,q} \cong \mathbb{R}$" [SS, §2.2]. Mochizuki's position is not to use it as an identification (Section 6); in the model, whether it is used can be read off from the indices. For a prime $l \geq 5$ we have $k - 1 = (l+4)(l-3)/12 > 0$. Finer normalizations (such as the $2l$-th root of q) are omitted in the model. This is the same model as [Ur, Setup 5.1].

5.2 The three options of Scholze–Stix

Proposition 5.3. P R

Known mathematics: [SS, §2.2]

In Setup 5.1:

(a) Use both. *If both ℓ and c are used as identifications (the diagram Γ_{two}) and values are required to be consistent, the only value is 0. Since $a = -Q \neq 0$, this requirement is incompatible with the Θ-link.*

(b) Make them one. *If ℓ and c are made the same identification ($\lambda_\ell = \lambda_c$), then $k = 1$: the factor j^2 is omitted. The comparison $a \leq c(b) + E$ then reads $-Q \leq -Q + E$, that is $0 \leq E$, and says nothing.*

(c) Blur. *If k is kept and the results of transport along the two paths are compared up to an error E, that is, if one asks for $\ell(b) \leq_E c(b)$, this is equivalent to $(k-1)Q \leq E$. Here $k - 1 = (l+4)(l-3)/12$ grows like l^2.*

Proof. (a) The closed path of Γ_{two} has monodromy $\lambda_c^{-1}\lambda_\ell = 1/k \neq 1$; apply Proposition 4.2 (ii). (b) $1/k = \lambda_\ell = \lambda_c = 1$ gives $k = 1$; then $b = -Q$ and $c(b) = -Q$. (c) Since $\ell(b) = a = -Q$ and $c(b) = -kQ$, the inequality $-Q \leq -kQ + E$ is equivalent to $(k-1)Q \leq E$. □

S (a) "However, it is clear that this will result in the whole diagram having monodromy j^2, i.e., being inconsistent."

(b) "The conclusion of this discussion is that with consistent identifications of copies of real numbers, one must ... omit the scalars j^2 that appear, which leads to an empty inequality."

(c) "In particular, he claimed that up to the 'blurring' given by certain indeterminacies the diagram does commute; it seems to us that this statement means that the blurring must be by a factor of at least $\mathrm{O}(\ell^2)$ rendering the inequality thus obtained useless."

[SS, §2.2]; the labels (a), (b), (c), showing the correspondence with Proposition 5.3, are ours

H The factor of at least $\mathrm{O}(\ell^2)$ in (c) corresponds in the model to the factor $k = l(l+1)/12$ that moves $b = -kQ$ to $a = -Q$, that is, to the added error $(k-1)Q$.

Companion paper: [Ur, Thms. 5.2 and 5.3]

5.3 The part that is not disputed

S (AD) "The correspondence that appears in the **Θ-link**

$$\{\underline{\underline{q}}^{j^2}\} \quad \mapsto \quad \underline{\underline{q}}$$

(where $j = 1, \ldots, l^*$) is, when considered within/relative to a **single hol. str.**, i.e., a **single ring/scheme theory**, manifestly **incompatible/inconsistent** with computing **arithmetic degrees**, unless one considers arithmetic degrees up to the **indeterminacy** *given by possible multiplication by factors 'j^2'* (where $j = 1, \ldots, l^*$) — an indeterminacy which then renders the computation *meaningless*."

(IUAD) "There is **absolutely no dispute** whatsoever (between SS and HM) with respect to the validity of the assertion (AD)."

[R18, §12, (AD), (IUAD)]

H Proposition 5.3 (a) corresponds to the first half of (AD) (inside a single ring theory the two are incompatible), and Proposition 4.3 to the second half (introducing the indeterminacy makes the computation meaningless). On this part there is, as Mochizuki says, no dispute.

6 Mochizuki's elementary numerical model

6.1 Example 3.11.4

S "the overall general *logical flow* of inter-universal Teichmüller theory — i.e., starting from the definition of the Θ-*link*, proceeding to the *multiradial representation of the* Θ-*pilot* [cf. [IUTchIII], Theorem 3.11], and finally, culminating in a *final numerical estimate* [cf. [IUTchIII], Corollary 3.12] — may be represented by means of *real numbers* $A, B \in \mathbb{R}_{>0}$ and $\epsilon, N \in \mathbb{R}$ such that $0 \leq \epsilon < 1$ in the following way:

- **Θ-link**:
$$\left(N \stackrel{\text{def}}{=} -2B\right) \quad \wedge \quad \left(N \stackrel{\text{def}}{=} -A\right);$$
- **multiradial representation of the Θ-pilot**:
$$\left(N = -2A + \epsilon\right) \quad \wedge \quad \left(N = -A\right);$$
- **final numerical estimate**:
$$-2A + \epsilon = -A, \text{ hence } A = \epsilon, \text{ i.e., } A < 1.$$

> Thus, the Θ-*link* ... and *multiradial representation of the* Θ-*pilot* ... are *meaningful/nontrivial* precisely because of the **logical relator** '$\wedge$', whose use obligates one, in the definition of the Θ-link, to consider *a priori* **distinct** real numbers A, B" ...
>
> "the use of *distinct* real numbers A, B in the definition of the '$\vee$' version of Θ-link seems *entirely superfluous* .... This motivates one to *identify* A and B — i.e., to suppose *'for the sake of simplicity'* that $A = B$ — which then has the effect of rendering the definition of the original '$\wedge$' version of the Θ-link *invalid/self-contradictory*"
>
> [Alien, Example 3.11.4]

6.2 Reading it with indices

H The ingredients of Example 3.11.4 correspond to the notation of Setup 5.1 as follows.

Example 3.11.4	This note (Setup 5.1)
A priori distinct positive reals A, B	The same size measured in the units of the q side (‡) and of the Θ side (†); without c the two cannot be compared
The factor 2	The mean k of j^2
$(N \stackrel{\text{def}}{=} -2B) \wedge (N \stackrel{\text{def}}{=} -A)$ (the Θ-link)	$b \approx_\ell a$: one object is described as b on the Θ side (in the example $-2B$) and as $a = -A$ on the q side
Identifying $A = B$ "for simplicity"	Using c as an identification as it stands and collecting everything into one number N; by Proposition 5.3 (a) the only value is 0 (a self-contradiction)
$(N = -2A + \epsilon) \wedge (N = -A)$ (the multiradial representation)	$a = c(b) + \varepsilon$: the results of transport along the two paths agree up to the error ε
$A = \epsilon$, $A < 1$ (the final estimate)	$(k-1)Q = \varepsilon$, $Q < 1/(k-1)$

Proposition 6.1. P R

Known mathematics: [Alien, Example 3.11.4]

In Setup 5.1, assume $k > 1$.

(i) If $a = c(b) + \varepsilon$ *with* $0 \le \varepsilon < 1$, *then* $(k-1)Q = \varepsilon$, *and hence* $Q < 1/(k-1)$. *For* $k = 2$ *this gives* $Q = \varepsilon < 1$, *the same formula as the final estimate* $A = \epsilon < 1$ *of Example 3.11.4.*

(ii) $\ell(b) \le_E c(b)$ *is equivalent to* $(k-1)Q \le E$.

Proof. Substitute $\ell(b) = a = -Q$ and $c(b) = -kQ$: (i) $-Q = -kQ + \varepsilon$; (ii) Proposition 5.3 (c). □

6.3 The content lies in the license

S "Put another way, this algorithm allows one to" ...

"(4^{cld}) construct a **closed loop** — i.e., from a given *input data* [*q-intertwined*!] $\mathcal{F}^{\mid\vdash\blacktriangleright\times\mu}$-prime-strip back to the given *input data* [*q-intertwined*!] $\mathcal{F}^{\mid\vdash\blacktriangleright\times\mu}$-prime-strip — whose *output data* consists of the Θ-*intertwining* [up to suitable *indeterminacies*] on this single given input data [*q-intertwined*!] $\mathcal{F}^{\mid\vdash\blacktriangleright\times\mu}$-prime-strip [cf. the discussion of the final portion of [IUTchIII], Remark 3.9.5, (ix)]. Here, we note that it is precisely this *closed* nature of the loop that allows one to derive ... *nontrivial consequences* ... from the *multiradial representation of the* Θ-*pilot*."

[Alien, §3.11 (v)]

H In the model this closed path goes from ‡ back to † along c reversed and returns to ‡ along ℓ. Its monodromy is $1/k$, not 1 (Proposition 5.3 (a)). That the output data is obtained

"up to suitable *indeterminacies*" can be read as saying that the results of transport along the two paths agree within the error E, that is, $\ell(b) \leq_E c(b)$.

Remark 6.2 (What is disputed). H In Example 3.11.4 the two descriptions determine $\epsilon = (k-1)A$ exactly ($\epsilon = A$ for $k = 2$). So "ϵ is small" and "A is small" are the same statement. When Mochizuki writes that, once one is equipped with this license, the derivation of the bounds on heights is an essentially formal matter, this is accurate in that sense: the content lies in the license itself.

What is disputed is whether Theorem 3.11 of IUT III (the multiradial algorithm) provides the license with a small error independently of the conclusion. Mochizuki says that it does; Scholze–Stix say that the blurring must be by a factor of at least $O(l^2)$, which renders the inequality thus obtained useless. The model is consistent with both: one can build models with $E \geq (k-1)Q$ and models with $E < (k-1)Q$. Which holds for the actual construction is outside the model.

Companion paper: [Ur, Thms. 5.3 and 5.7]

7 Corollary 3.12 and the upper bound of IV

> S Corollary 3.12: write $-|\log(\underline{\underline{\Theta}})| \in \mathbb{R} \cup \{+\infty\}$ for the procession-normalized mono-analytic log-volume of the holomorphic hull of the union of the possible images of a Θ-pilot object in the multiradial representation of Theorem 3.11, which is regarded as subject to the indeterminacies (Ind1), (Ind2), (Ind3); and write $-|\log(\underline{\underline{q}})| \in \mathbb{R}$ for the procession-normalized mono-analytic log-volume of the image of a q-pilot object, which is not regarded as subject to these indeterminacies. Here $|\log(\underline{\underline{q}})| > 0$. Then "it holds that $-|\log(\underline{\underline{\Theta}})| \in \mathbb{R}$, and $-|\log(\underline{\underline{\Theta}})| \geq -|\log(\underline{\underline{q}})|$".
>
> [IUT3, Cor. 3.12]

Proposition 7.1. P R

Known mathematics: [SS, §2.2]

In Setup 5.1, let $-|\log\Theta| := c(b) + E$ be the value of the hull of the possible images of the Θ-pilot, computed through c. Then the inequality $-|\log\Theta| \geq -|\log q|$ of Corollary 3.12 is equivalent to $\ell(b) \leq_E c(b)$, and hence to $(k-1)Q \leq E$.

Proof. Since $-|\log q| = a = \ell(b)$, the inequality reads $\ell(b) \leq c(b) + E$; now apply Proposition 6.1 (ii). □

Corollary 7.2. P R

Known mathematics: [Alien, Introduction], [IUT4, Thm. 1.10]

If IV bounds the error from above by $E \leq C$, with C independent of Q, then Corollary 3.12 implies $Q \leq C/(k-1)$.

Proof. $(k-1)Q \leq E \leq C$ and $k - 1 > 0$. □

Remark 7.3. H (i) For $k = 1$ (omitting j^2) we get $k - 1 = 0$ and nothing follows. The information, an upper bound for Q, comes from $k > 1$, that is, from j^2; and $k > 1$ is also what produces the monodromy $1/k \neq 1$ of the two paths. The content of Corollary 3.12 and the difficulty of the dispute come from the same j^2.

(ii) The actual estimate of IV [IUT4, Thm. 1.10] contains terms such as differents and conductors, and C depends on l. The model keeps only the shape. Q is the size of q, and its upper bound has the shape of a height inequality (an inequality of abc type).

Companion paper: [Ur, Cor. 5.4]

8 The two positions in one table

H The following table summarizes the above in the notation of Setup 5.1.

	Scholze–Stix [SS, §2.2]	Mochizuki [R18, Alien]	Common ground
Identifications used	Both ℓ and c, consistently	Only the Θ-link ℓ; c is recovered with an error by the multiradial algorithm	Using both ℓ and c as they stand leads to a contradiction ((AD), (IUAD); Prop. 5.3 (a))
Graph	Γ_{two} (with a cycle)	Γ_{seg} (a segment); Γ_{uni} if the number of turns is to be distinguished	If the monodromy of a cycle is not 1, the only consistent value is 0 (Prop. 4.2)
Reading of Cor. 3.12	Omitting j^2 gives $0 \leq E$, which is empty; keeping it needs an error of at least $(k-1)Q$, which is useless	From the license with a small error, the height bound follows formally	In the model, Cor. 3.12 $\Leftrightarrow (k-1)Q \leq E$ (Prop. 7.1)
The decisive point	The error cannot be made small	Theorem 3.11 provides a small error	The decisive point is the size of the error E (its value lies outside the model)

9 One level down: presented reals

S "The theorem is proved over *presented* reals—regular sequences of rationals with Bishop equality—... rather than for an abstract real-number object."

[U, Abstract]

In his formalization of presented reals [U], the author treated real numbers through presentations that carry witnesses of convergence. Two presentations of the same real number are related by a proposition, namely whether they are equal as real numbers. H In the language of this note, in the container reached after forgetting presentations, the scale of every identification is 1. Then every closed path has monodromy 1, and the obstruction of Proposition 4.2 (ii) does not occur (Exercise 9). Forgetting a presentation loses only the presentation, and a forgotten witness can be recovered without any choice principle when there is a decidable search.

S "In Lean, a statement in `Prop` is proof-irrelevant, and an arbitrary witness hidden in a `Prop`-valued existential cannot in general be eliminated into `Type` (data can nevertheless be recomputed when an independent decidable search produces a witness—Appendix B.12)"

[U, §7.2]

H Presented reals correspond to the case in which all scales are 1. The difficulty of IUT lies in the fact that an identification of scale $1/k \neq 1$ cannot be avoided. The relation with locators, and a formulation that treats both levels in the single language of homotopy type theory, are left to [Ur, §7].

Part II
Components

10 What one number cannot hold

In Part I the error E was a single number, and the dispute became the inequality $(k-1)Q \leq E$ (Proposition 7.1). In the source, E is not a free parameter. It comes from the indeterminacies

(Ind1), (Ind2), (Ind3) acting on the regions of the multiradial representation, and these regions live in *tensor packets*.

> S "we consider, for each element $* \in \mathbb{S}^{\pm}_{j+1}$, and for each [say, for simplicity, *nonarchimedean*] $v_{\mathbb{Q}} \in \mathbb{V}_{\mathbb{Q}}$,
>
> · the *direct sum of the* **log-shells** associated to the prime-strip labeled by the given element $* \in \mathbb{S}^{\pm}_{j+1}$ at the $\underline{v} \in \underline{\mathbb{V}}$ that lie over $v_{\mathbb{Q}}$;
>
> we then form
>
> · the **tensor product**, over the elements $* \in \mathbb{S}^{\pm}_{j+1}$, of these *direct sums*.
>
> This collection of tensor products associated to $v_{\mathbb{Q}} \in \mathbb{V}_{\mathbb{Q}}$ will be referred to as the **tensor packet** associated to the collection of prime-strips indexed by elements of $\mathbb{S}^{\pm}_{j+1}$."
>
> [IUT3, Introduction]

When the dispute is written in a proof assistant, the regions and the indeterminacies have to be represented. The public Lean formalization [LANA-iut] does this with tuples of places, their weights and the permutations of the labels, and it distinguishes two readings of the union in Corollary 3.12.

> S "The indeterminacy (Ind1) contains the permutation automorphisms of the label sets S_{j+1} of the procession (IUT III, Introduction): the theta value q^{j^2} of the capsule S_{j+1} may sit at any label. IUT IV, proof of Theorem 1.10, Steps (iv)–(v), computes with the theta value at the label j only."
>
> [LANA-iut, `Iut/Concrete/Literal.lean`]; code markup rendered as mathematics

Read backwards from such a formalization, the notation of Section 3 lacks four things.

(1) *Values on components.* A value is not one number but a family of numbers, one for each component of a tensor packet.

(2) *Indices that are pairs.* An identification is a permutation of the components together with an identification on each component; composing two of them is composition in a wreath product.

(3) *An order for collecting possible images.* The hull of a union, which takes at each component the largest possible image, and the average over the components do not commute.

(4) *Inclusions and provenance.* Regions may be enlarged, which is what (Ind3) allows, and a number that is compared has to carry the construction it comes from.

Section 11 adds these to the notation. Section 12 builds a model with components, Section 13 computes the hull in two readings of the union, and Section 14 writes the dispute in the extended notation.

11 Indexed identifications with components

Definition 11.1 (Components and values)**.** R

Known mathematics: [IUT3, Introduction] (tensor packets), [LANA-iut]

Let C be a finite set, the *components*, with weights $w(c) \geq 0$. A *value* on C is a function $x : C \to \mathbb{R}$; we think of $x(c)$ as the log-volume of a region in the component c. For two values we write $x \sqsubseteq y$ if $x(c) \leq y(c)$ for every c (the region of x is contained in that of y). The *average* of x is $\mu(x) := \sum_{c \in C} w(c)\, x(c)$.

H In the model the regions in one component are nested, like balls with a common centre, so a region is determined by its log-volume, inclusion is the order of log-volumes, and the union of finitely many regions is the largest of them. This simplification is what makes a hull a maximum.

Definition 11.2 (Identifications with components)**.** R

Known mathematics: wreath products of groups; Definition 3.2 for one component

An *identification with components* is a pair (σ, f) of a permutation σ of C with $w(\sigma c) = w(c)$ for all c and a family $f = (f_c)_{c\in C}$ of identifications in the sense of Definition 3.2, one for each component. It acts on values by

$$((\sigma, f) \cdot x)(c) := \lambda_{f_c}\, x(\sigma^{-1}c),$$

and we write $x \approx_{(\sigma,f)} y$ for $y = (\sigma, f) \cdot x$.

Proposition 11.3 (Rules with components)**.** P R

Known mathematics: wreath products; for one component and $\sigma = \mathrm{id}$, Proposition 3.3

(R1$^+$) Identity and inverse. $x \approx_{(\mathrm{id},\mathrm{id})} x$. *If* $x \approx_{(\sigma,f)} y$, *then* $y \approx_{(\sigma^{-1},f')} x$, *where* $\lambda_{f'_c} = 1/\lambda_{f_{\sigma c}}$.

(R2$^+$) Composition. *If* $x \approx_{(\sigma,f)} y$ *and* $y \approx_{(\tau,g)} z$, *then* $x \approx_{(\tau\sigma,h)} z$ *with* $\lambda_{h_c} = \lambda_{g_c}\, \lambda_{f_{\tau^{-1}c}}$.

(R3$^+$) Action on numbers. *Values change componentwise as in (R3). If every* f_c *has scale* 1, *the average is an invariant:* $\mu((\sigma, f) \cdot x) = \mu(x)$.

(R4$^+$) Sets of indices, hulls and averages. *For a finite set* S *of identifications with components, the possible images* $S(x)$ *are as in (R4). The* hull $H_S(x)(c) := \max_{s\in S}(s \cdot x)(c)$ *is the least value* y *with* $s \cdot x \sqsubseteq y$ *for all* $s \in S$, *and* $\mu(H_S(x)) \geq \mu(s \cdot x)$ *for every* $s \in S$.

(R5$^+$) Inclusions. *If* $x \sqsubseteq y$, *then* $\mu(x) \leq \mu(y)$. *If* $S \subseteq S'$, *then* $H_S(x) \sqsubseteq H_{S'}(x)$.

Proof. For (R1$^+$) and (R2$^+$), compute $((\tau, g) \cdot ((\sigma, f) \cdot x))(c) = \lambda_{g_c}\lambda_{f_{\tau^{-1}c}}\, x(\sigma^{-1}\tau^{-1}c)$ and use $(\tau\sigma)^{-1} = \sigma^{-1}\tau^{-1}$; the inverse is the pair whose composite with (σ, f) is $(\mathrm{id}, \mathrm{id})$. For (R3$^+$), substitute $c = \sigma c'$ in $\sum_c w(c)\, x(\sigma^{-1}c)$ and use $w(\sigma c') = w(c')$. (R4$^+$) and (R5$^+$) hold because a maximum is the least upper bound and the weights are nonnegative. □

Definition 11.4 (Provenance)**.** R A number of the form $\mu(H_S(x))$ is recorded together with S and x. Two inequalities may be chained through such numbers only if they refer to the same record, or if (R5$^+$) gives the inequality between the two numbers that the chaining needs.

Remark 11.5. H Compared with Section 3, (R1) and (R3) hold componentwise, (R2) becomes composition in a wreath product, and (R4) acquires the hull and the average. These two do not commute: by (R3$^+$) every possible image has the same average, while the average of the hull is in general larger (Section 13). This is the precise sense in which "taking the union" and "measuring" must be done in a declared order. The Θ-link ℓ of Part I relates the Θ-pilot to the q-pilot, whose log-volume is the single number $a = -Q$; in Part II only the identifications on the Θ side and the natural identification c, of scale 1 on every component, are used.

12 The model with components

S "where we write $\mathbb{S}^{\pm}_{j+1} \overset{\text{def}}{=} \{0, 1, \ldots, j\}$, for $j = 0, \ldots, l^*$, and we think of each of these finite sets as being subject to arbitrary permutation automorphisms. That is to say, we think of the set $\mathbb{S}^{\pm}_{j+1}$ as a **container** for the labels $0, 1, \ldots, j$. Thus, for each j, one need only contend with an *indeterminacy of* $j + 1$ *possibilities* for the element of this container that corresponds to j." ...

"It turns out that this reduction will yield just the right estimates in the log-volume computations to be performed in [IUTchIV]."

[IUT3, Introduction]

S Step (iv): "In particular, for *each collection* $\{v_i\}_{i\in\mathbb{S}^{\pm}_{j+1}}$ of [not necessarily distinct!] elements of $\mathbb{V}(F_{\mathrm{mod}})_{v_{\mathbb{Q}}}$, we must estimate the component of the log-volume in question corresponding to the *tensor product* of the $\mathbb{Q}$-spans of the *log-shells* associated to this collection $\{v_i\}_{i\in\mathbb{S}^{\pm}_{j+1}}$ and then compute the *weighted average* [cf. the discussion of Remark 1.7.1], over possible collections $\{v_i\}_{i\in\mathbb{S}^{\pm}_{j+1}}$, of these estimates."

Step (v), applying Proposition 1.4 (iii) of IUT IV, takes "$i^{\dagger}$" to be j and "λ" to be the order of the theta value $\underline{\underline{q_v^{j^2}}}_j$ at the place of the factor j (0 if this place is good). It continues: "That is to say, the *indeterminacies* (Ind1) and (Ind2) are taken into account by the *arbitrary* nature of the automorphism "ϕ" [cf. Proposition 1.2]" ...

"Also, we note that, unlike the other terms that appear in these *upper bounds*, "λ" is *asymmetric* with respect to the choice of "$i^{\dagger}\in I$" in $\mathbb{S}^{\pm}_{j+1}$." ...

"it follows immediately from the first equality of the first display of Proposition 1.7 that, after passing to *weighted averages*, the operation of *symmetrizing* with respect to the choice of "$i^{\dagger}\in I$" in $\mathbb{S}^{\pm}_{j+1}$ does *not affect* the computation of the *upper bound* under consideration."

[IUT4, proof of Thm. 1.10, Steps (iv), (v)]

Setup 12.1 (The model with components). R

Known mathematics: [IUT3, Introduction], [IUT4, proof of Thm. 1.10, Steps (iv), (v)], [LANA-iut]

Consider the following data.

(a) *One prime and its places.* A finite set V of places above one prime, with weights $w_v>0$, $\sum_v w_v=1$, and q-orders $Q_v\geq 0$; $Q_v=0$ at a good place and $Q_v>0$ at a bad place. Put $Q:=\sum_v w_vQ_v$; as in Setup 5.1, $a=-Q$.

(b) *Levels, factors and components.* For each level $j=1,\dots,l^*$ the factors are indexed by $\mathbb{S}_{j+1}=\{0,1,\dots,j\}$, and the components of level j are the tuples $t=(t_0,\dots,t_j)\in V^{j+1}$, with weight $w(t):=w_{t_0}w_{t_1}\cdots w_{t_j}$.

(c) *The Θ-pilot.* $b=(b_j)_j$ with $b_j(t):=-j^2Q_{t_j}$: the theta value q^{j^2} sits in the factor j.

(d) *Label permutations.* A permutation τ of $\mathbb{S}_{j+1}$ acts on the components of level j by $t\mapsto t\circ\tau^{-1}$, which preserves the weights, and on values by $(\tau\cdot x)(t):=x(t\circ\tau)$. Thus $(\tau\cdot b_j)(t)=-j^2Q_{t_{\tau(j)}}$: the theta value now sits in the factor $\tau(j)$.

(e) *Procession-normalized values.* For a family $x=(x_j)_j$ put

$$\bar{\mu}(x):=\frac{1}{l^*}\sum_{j=1}^{l^*}\sum_{t\in V^{j+1}}w(t)\,x_j(t).$$

Then $\bar{\mu}(b)=-kQ$, with k the mean of j^2 as in Setup 5.1 (Exercise 10).

(f) *The rest of the blur.* The identifications on the components have scale 1, as c has in Setup 5.1. What (Ind2) and (Ind3) add to a hull is represented, as in Part I, by one number $E\geq 0$, and $C\geq 0$ is the constant of the upper bound of IV; neither depends on the Q_v.

Definition 12.2 (Readings of the union). R

Known mathematics: [LANA-iut] (`litLabels`, `ivLabels`)

A *reading* is a family $S=(S_j)_j$ of sets S_j of permutations of $\mathbb{S}_{j+1}$, each containing the identity. Its hull is $H_S(b)_j(t):=\max_{\tau\in S_j}(\tau\cdot b_j)(t)$, and its value is $M(S):=\bar{\mu}(H_S(b))$. We use:

- S_{a}: all permutations of $\mathbb{S}_{j+1}$, so that the element labelled j may sit in any of the $j+1$ places;
- S_{b}: the permutations with $\tau(j)=j$, so that the theta value stays in the factor j;
- the *symmetrization* $b_j^{\mathrm{sym}}(t) := \frac{1}{j+1}\sum_{i=0}^{j}(-j^2 Q_{t_i})$, the average at each component of the $j+1$ values that the theta value can give.

Remark 12.3. H In Setup 12.1, (b) is the structure of the tensor packets quoted above: a tensor product over $\mathbb{S}_{j+1}$ of direct sums over the places, decomposed into components indexed by collections $\{v_i\}_{i\in\mathbb{S}_{j+1}}$; (c) is the choice $i^\dagger = j$ in Step (v) of IV; and the sum in (e) is the weighted average of Step (iv) followed by the average over j. The reading S_{a} is the one in which the $j+1$ possibilities of the introduction of IUT III enter the union of possible images; S_{b} is the computation of Step (v), and b^{sym} is its symmetrization over $i^\dagger$. The two readings are those called the literal reading and IV's reading in [LANA-iut]. The normalizations of IV (the factors $1/(j+1)$ and $1/2l$) and the differents are omitted.

13 The hull in the two readings

Proposition 13.1. P R

Known mathematics: elementary; the qualitative content is in [LANA-iut]; for (iv), [IUT4, Prop. 1.7], formalized in [LANA-iut]

In Setup 12.1, for every level j and every component t:

(i) $H_{S_{\mathrm{a}}}(b)_j(t) = -j^2 \min_{0\le i\le j} Q_{t_i}$*: the hull is governed by the smallest q-order among the factors of t.*

(ii) $H_{S_{\mathrm{b}}}(b) = b$.

(iii) $b_j^{\mathrm{sym}}(t) \le H_{S_{\mathrm{a}}}(b)_j(t)$*: where the hull takes the largest of the possible values, the symmetrization takes their average.*

(iv) For every permutation $\tau=(\tau_j)_j$, $\bar\mu(\tau\cdot b)=\bar\mu(b)$; and $\bar\mu(b^{\mathrm{sym}})=\bar\mu(b)=-kQ$.

Proof. (i) For each $i\in\mathbb{S}_{j+1}$ the transposition of j and i sends j to i, so the possible values at t are exactly $-j^2Q_{t_i}$ for $i=0,\dots,j$, and the largest is $-j^2\min_i Q_{t_i}$. (ii) If $\tau(j)=j$, then $(\tau\cdot b_j)(t)=b_j(t)$. (iii) An average is at most the maximum. (iv) The map $t\mapsto t\circ\tau$ is a bijection of V^{j+1} that preserves w (R3$^+$). For each i we have $\sum_t w(t)\,Q_{t_i}=\sum_v w_v Q_v = Q$, because the factor i of a tuple is drawn from w independently of the others (Exercise 10); hence the average of b_j^{sym} is $-j^2Q$, the same as that of b_j. □

Proposition 13.2 (The difference is an order statistic)**.** P R

Known mathematics: elementary probability

For $n\ge 1$ let $m_n := \sum_{t\in V^n} w(t)\min_i Q_{t_i}$, the expected value of the smallest of n independent draws of the q-order, a place v being drawn with probability w_v. Then

$$M(S_{\mathrm{a}}) - M(S_{\mathrm{b}}) = \frac{1}{l^*}\sum_{j=1}^{l^*} j^2\,(Q - m_{j+1}) \;\ge\; 0,$$

and the difference is 0 if and only if all places have the same q-order.

Proof. By Proposition 13.1 (i), $M(S_{\mathrm{a}}) = -\frac{1}{l^*}\sum_j j^2 m_{j+1}$; by (ii) and (iv), $M(S_{\mathrm{b}}) = -kQ = -\frac{1}{l^*}\sum_j j^2 Q$. Since $\min_i Q_{t_i}\le Q_{t_j}$, we have $m_{j+1}\le Q$. If all Q_v are equal, then $\min_i Q_{t_i}=Q_{t_j}$ for every t. If $Q_u<Q_v$ for two places, the tuples with $t_0=u$ and $t_j=v$ have positive weight and satisfy $\min_i Q_{t_i}<Q_{t_j}$ (recall $j\ge 1$), so $m_{j+1}<Q$. □

Proposition 13.3 (Two places). P *(elementary)*

Let V consist of a bad place with weight $w \in (0,1)$ and q-order $Q_{\mathrm{b}} > 0$ and a good place with weight $1-w$. Then $Q = wQ_{\mathrm{b}}$ and $m_{j+1} = w^{j+1}Q_{\mathrm{b}}$; at a component, $H_{S_{\mathrm{a}}}(b)_j(t)$ is $-j^2Q_{\mathrm{b}}$ if every factor of t sits at the bad place and 0 otherwise. Hence

$$M(S_{\mathrm{a}}) = -Q\,A(w), \qquad A(w) := \frac{1}{l^*}\sum_{j=1}^{l^*} j^2 w^j, \qquad M(S_{\mathrm{b}}) = -kQ.$$

Proof. The smallest q-order among the factors is Q_{b} if all factors sit at the bad place, which happens with weight w^{j+1}, and 0 otherwise. Hence $M(S_{\mathrm{a}}) = -\frac{1}{l^*}\sum_j j^2 w^{j+1}Q_{\mathrm{b}} = -wQ_{\mathrm{b}} \cdot \frac{1}{l^*}\sum_j j^2 w^j$. □

S "The two regions coincide when $\mathrm{ord}_p(q_v)$ is the same for all places v of F_{mod} (in V) above each prime p (e.g. when there is a single place above p): then the theta values at all labels of a tuple have the same order. This fails in general; two bad places above p with different $\mathrm{ord}_p(q_v)$ already break it."

"Under the literal union over the labels, the holomorphic hull at a tuple c is governed by $\min_l \mathrm{ord}_p(q_{c(l)})$ rather than by the average of $\mathrm{ord}_p(q_v)$ over the places that IUT IV, Step (v), obtains via Proposition 1.7. So the literal reading does not suffice for the Theorem 1.10 / ABC argument: the points produced by the Belyi reduction have good and bad places above the same prime, where $\min_l \mathrm{ord}_p(q_{c(l)}) = 0$."

[LANA-iut, `Iut/Concrete/Literal.lean`]; code markup rendered as mathematics

H Proposition 13.1 (i) is the minimum of this description, Proposition 13.2 says when the two readings agree, and Proposition 13.3 is the situation with good and bad places above one prime. What the model adds is the size of the difference: an expected order statistic, which for two places equals $(k - A(w))\,Q$.

14 The dispute with components

Definition 14.1. R

Known mathematics: [IUT3, Cor. 3.12], [IUT4, Thm. 1.10]

For a reading S we write

$$\mathrm{Cor}(S): \quad -Q \le M(S) + E, \qquad\qquad \mathrm{IV}(S): \quad M(S) + E \le -kQ + C.$$

Lemma 14.2. P *(elementary)*

For every $n \ge 1$, $\sum_{j=1}^{n} j^2\,2^{-j} = 6 - (n^2+4n+6)/2^n \le n$. Hence $A(w) \le A(\frac{1}{2}) \le 1$ for $0 \le w \le \frac{1}{2}$ and every $l^* \ge 1$.

Proof. The identity follows by induction on n. For $n \ge 6$ the sum is less than $6 \le n$; for $n = 1, \dots, 5$ it is $\frac{1}{2}, \frac{3}{2}, \frac{21}{8}, \frac{29}{8}, \frac{141}{32}$. Finally, A is increasing on $w \ge 0$. □

Theorem 14.3. P R

Known mathematics: elementary; for (i), [LANA-iut] (`cor312Literal_of_variant`)

In Setup 12.1:

(i) Monotonicity. *If $S_j \subseteq S'_j$ for all j, then $M(S) \le M(S')$. Hence* $\mathrm{Cor}(S)$ *implies* $\mathrm{Cor}(S')$, *and* $\mathrm{IV}(S')$ *implies* $\mathrm{IV}(S)$. *In particular* $\mathrm{Cor}(S_{\mathrm{b}}) \Rightarrow \mathrm{Cor}(S_{\mathrm{a}})$ *and* $\mathrm{IV}(S_{\mathrm{a}}) \Rightarrow \mathrm{IV}(S_{\mathrm{b}})$.

(ii) Conservation. *For every reading S,* $\mathrm{Cor}(S)$ *and* $\mathrm{IV}(S)$ *together imply $(k-1)Q \le C$.*

(iii) The reading S_{b}. $\mathrm{IV}(S_{\mathrm{b}}) \Leftrightarrow E \le C$, *and* $\mathrm{Cor}(S_{\mathrm{b}}) \Leftrightarrow (k-1)Q \le E$.

(iv) The reading S_{a}, with two places as in Proposition 13.3. *If* $A(w) \le 1$, *in particular if* $w \le \frac{1}{2}$, *then* $\mathrm{Cor}(S_{\mathrm{a}})$ *holds for every* $Q_{\mathrm{b}} > 0$ *and every* $E \ge 0$; *for* $E = 0$, $\mathrm{Cor}(S_{\mathrm{a}})$ *holds if and only if* $A(w) \le 1$. *Since* $A(w) < k$, $\mathrm{IV}(S_{\mathrm{a}})$ *fails as soon as* $Q_{\mathrm{b}} > (C - E)/(w(k - A(w)))$.

(v) Replacing q by q^N. *In the situation of (iv), replace* Q_{b} *by* NQ_{b} *with* $N > 0$, *and let* $l^* \ge 2$, *so that* $k > 1$. *If* $A(w) \le 1$, $\mathrm{Cor}(S_{\mathrm{a}})$ *holds for every* N, *while* $\mathrm{Cor}(S_{\mathrm{b}})$ *fails as soon as* $N > E/((k-1)wQ_{\mathrm{b}})$.

Proof. (i) By $(\mathrm{R5}^+)$, $H_S(b) \sqsubseteq H_{S'}(b)$, so $M(S) \le M(S')$; then $-Q \le M(S) + E \le M(S') + E$, and $M(S) + E \le M(S') + E \le -kQ + C$. (ii) $-Q \le M(S) + E \le -kQ + C$. (iii) $M(S_{\mathrm{b}}) = -kQ$ by Proposition 13.1 (ii) and (iv). (iv) By Proposition 13.3, $\mathrm{Cor}(S_{\mathrm{a}})$ reads $-Q \le -QA(w) + E$; this holds if $A(w) \le 1$ and, for $E = 0$, only then. Lemma 14.2 gives $A(w) \le 1$ for $w \le \frac{1}{2}$. $\mathrm{IV}(S_{\mathrm{a}})$ reads $-QA(w) + E \le -kQ + C$, that is $wQ_{\mathrm{b}}\,(k - A(w)) \le C - E$; and $A(w) < k$ because $w^j < 1$. (v) The condition $A(w) \le 1$ does not involve Q_{b}, and by (iii) $\mathrm{Cor}(S_{\mathrm{b}})$ reads $(k-1)NwQ_{\mathrm{b}} \le E$. □

Corollary 14.4 (A mixed pair says nothing). P R
Known mathematics: elementary

In the situation of Theorem 14.3 (iv) with $w \le \frac{1}{2}$ *and* $E \le C$, *the statements* $\mathrm{Cor}(S_{\mathrm{a}})$ *and* $\mathrm{IV}(S_{\mathrm{b}})$ *hold together for every* $Q_{\mathrm{b}} > 0$. *They give no bound on* Q.

Proof. Theorem 14.3 (iv) and (iii). □

Remark 14.5 (Why provenance matters). H If the two hulls $M(S_{\mathrm{a}})$ and $M(S_{\mathrm{b}})$ were treated as one number, the two statements of Corollary 14.4 would give $(k-1)Q \le C$ for every Q_{b}, which is false in the model as soon as $Q_{\mathrm{b}} > C/((k-1)w)$. The record of Definition 11.4 forbids this chaining. A pair of statements for one reading always contains a statement that carries the height bound (Theorem 14.3 (ii)): for S_{b} it is $\mathrm{Cor}(S_{\mathrm{b}})$, the inequality $(k-1)Q \le E$ of Part I; for S_{a} it is $\mathrm{IV}(S_{\mathrm{a}})$. The reading decides where the content sits, not whether it is there.

15 Correspondences and what remains

15.1 With Scholze–Stix

H In the reading S_{a} with $A(w) \le 1$, the label permutations alone produce a blur

$$M(S_{\mathrm{a}}) - M(S_{\mathrm{b}}) = (k - A(w))\,Q \;\ge\; (k-1)\,Q,$$

which is the size required in Proposition 5.3 (c), and Corollary 3.12 then holds whatever Q is (Theorem 14.3 (iv), (v)). This is the situation of option (c) of Scholze–Stix quoted in Section 5: a blurring by a factor of at least $O(\ell^2)$, after which the inequality says nothing about Q. In the reading S_{b} the label permutations blur nothing, and Corollary 3.12 is the inequality $(k-1)Q \le E$ of Part I with E coming from (Ind2) and (Ind3) alone. Scholze–Stix discuss neither label permutations nor places; the correspondence is one of shape.

15.2 With the formalizations

H The literal reading and IV's reading of [LANA-iut] are S_{a} and S_{b}, and the implication from IV's reading to the literal one, proved there, is Theorem 14.3 (i) in the model. The resolution adopted in [LANA-iut] is to take the hull for each (Ind1)-image separately.

> S "Hence the procession-normalized log-volume of the hull is (Ind1)-invariant" … "and every single-label reading gives IUT IV's value" … "IUT IV takes the holomorphic hull over (Ind2), (Ind3) for each (Ind1)-image separately, and the label j is a representative."
>
> [LANA-iut, `Iut/Concrete/Literal.lean`]

H In the model this is Proposition 13.1 (iv): every image $\tau \cdot b$ has the value $\bar{\mu}(b) = -kQ$, so measuring each image separately gives the value of S_{b}. Whether this is what Corollary 3.12 defines is the first point of Section 15.3.

That the actual (Ind1) includes permutations between tensor factors was noted earlier by Dupuy and Hilado.

> S "The only reason this subsection can't directly be applied is because the actual ind1 has some permutations among different tensor product factors of $\mathbb{A}_{\underline{V},p}^{\otimes j+1}$. The permutation of these factors does not appear in this example."
>
> [DH, §6.2, footnote 7]

In the setting of Part I, the inequality of Proposition 7.1 is stated in [TK] as the condition for the containment used in Step (v) of IV.

> S "the satisfiability of (P) for actual curves is logically equivalent to the ledger absorbing the Gauss gain, i.e. to $L \geq (l(l+1)/12 - 1) \cdot |\log(q)|$ — the height inequality itself."
>
> [TK, `REQUEST_theta_hull_derivation.md`]

An explicit initial Θ-datum, with two bad places of different q-orders above one prime, for which, under the readings stated there, the union in the literal reading is not contained in the module used in Step (v), is given in [Note, §§4–6].

15.3 What remains

H The extended notation does not decide the dispute. It separates two points.

(1) *Which reading is the source's.* Corollary 3.12 speaks of the hull of the union of the possible images under (Ind1), (Ind2), (Ind3) (Section 7), and the introduction of IUT III gives the element labelled j its $j+1$ possibilities; read literally, this is S_{a}. Step (v) of IV computes with $i^{\dagger} = j$ and symmetrizes over $i^{\dagger}$, which gives the value of S_{b}. Mochizuki writes that, at the bad places, (Ind1) and (Ind2) affect the log-volumes only to essentially negligible order:

> S (Ind3>1+2) "once one constructs the *output* of the **multiradial representation** of the Θ-pilot [cf. [IUTchIII], Theorem 3.11, (ii)] via **tensor-packets of log-shells** in such a way that each local portion of this output is *stable* with respect to the indeterminacy (Ind3), these local portions of the output are *automatically "essentially stable"* [i.e., stable up to discrepancies at the valuations $\in \underline{\mathbb{V}}^{\mathrm{bad}}$ that affect the resulting log-volumes only up to *very small/essentially negligible order*] with respect to the indeterminacies (Ind1), (Ind2) [cf. [IUTchIII], Theorem 3.11, (i)]."
>
> [Ess, §3.11]

In the model the discrepancy between the two readings is 0 at a prime above which all places have the same q-order, and otherwise an expected order statistic proportional to the q-orders (Proposition 13.2). The model does not decide which reading the source intends.

(2) *In the reading S_{b}, whether Theorem 3.11 gives the license*, that is, whether $(k-1)Q \leq E$ holds with E small enough for IV. This is the dispute of Part I, unchanged.

By Theorem 14.3 (ii), whichever reading is taken, Corollary 3.12 together with the upper bound of IV implies the height bound: in the reading S_{a} the content is carried by the upper bound of IV, in the reading S_{b} by Corollary 3.12.

15.4 Relation to the companion paper

H The companion paper [Ur] observes, at the end of its §4, that the result of descent changes according to over the orbits of which groupoid the union is taken. Part II gives two such groupoids concretely, the full symmetric groups of the $\mathbb{S}_{j+1}$ (the reading S_{a}) and the stabilizers of j (the reading S_{b}), and Proposition 13.2 computes the difference. The hull along the whole loop in [Ur], which is Proposition 4.3 here, concerns another orbit, that of the monodromy, which is infinite; the orbits of Part II are finite.

16 Summary

- Written with indexed identifications, "identify or not" becomes a formula about which edges of which diagram are used.
- *Agreed:* using consistently an identification that does not return to 1 around a loop leaves only the value 0 (Props. 4.2 and 5.3 (a); (AD), (IUAD)). If the labels are dropped and one may go around the loop any number of times, the comparison becomes empty (Prop. 4.3; [IUT3, Remark 3.11.1 (vii)]).
- *Disputed:* whether the comparison $\ell(b) \leq_E c(b)$ of the two paths holds with an error small enough to be compatible with the estimate $E \leq C$ of IV. In the model this is equivalent to $(k-1)Q \leq E$, which itself has the shape of an upper bound for Q (Prop. 7.1 and Cor. 7.2).
- The model does not decide this question. What decides it is the size of the error produced by the actual construction.
- *Part II.* With components, the error splits: the label permutations of (Ind1) produce a part that can be computed, and (Ind2), (Ind3) the rest. Whether the union of possible images includes the images with the theta value in another factor (S_{a}) or not (S_{b}) changes the hull by an expected order statistic, which is 0 when all places above the prime have the same q-order (Props. 13.1–13.3).
- For every reading, Corollary 3.12 and the upper bound of IV together imply the height bound; the reading decides only which of the two statements carries it (Thm. 14.3). A pair taken from two readings gives nothing (Cor. 14.4).

17 Exercises

Exercise 1. In the conversion table of Example 2.1, replace the last entry by $x_A = \frac{1}{6}x_C$ and find all consistent measurements. Explain, using Proposition 4.2, why with $\frac{1}{5}$ the only one is 0.

Exercise 2. Check, using (R1) and (R2), that the relation $x \approx y$ of Proposition 3.6 is an equivalence relation (reflexive, symmetric and transitive). How many equivalence classes are there?

Exercise 3. Using $\sum_{j=1}^{n} j^2 = n(n+1)(2n+1)/6$, show that for $l^* = (l-1)/2$ one has $k = l(l+1)/12$ and $k - 1 = (l+4)(l-3)/12$. Compute k for $l = 5, 7, 11, 13$.

Exercise 4. For the diagram on the universal cover Γ_{uni} with scale λ on every edge, show that the consistent families of values are exactly $x_n = \lambda^n x_0$.

Exercise 5. Check Proposition 4.3 for $m = \frac{1}{2}$ and $x_0 = -1$: list the elements of O for $n = -2, \ldots, 2$.

Exercise 6. From the computation $-2A + \epsilon = -A$ of Example 3.11.4, derive $A = \epsilon$. What do you get if the factor 2 is replaced by k? What if the equality is replaced by the inequality $-A \leq -kA + E$?

Exercise 7. In Corollary 7.2, find the upper bound for Q when $l = 7$ and $C = 1$. What happens when $k = 1$?

Exercise 8. Which of the following are unchanged by all identifications: the sign; the absolute value; the ratio of two values in the same container; the difference of two values in the same container?

Exercise 9 (One level down)**.** Show that in a connected diagram all of whose edges have scale 1, every closed path has monodromy 1, and that a consistent family of values can be built from any value at any vertex. This is the case of the presented reals of Section 9.

Exercise 10. In Setup 12.1, show that $\sum_{t \in V^{j+1}} w(t)\, Q_{t_i} = Q$ for every factor i, by writing the sum over t as $j+1$ sums over the entries. Deduce $\bar{\mu}(b) = -kQ$.

Exercise 11. Take $l = 5$, so $l^* = 2$, and two places as in Proposition 13.3. Compute $A(w)$ and find the largest w with $A(w) \leq 1$.

Exercise 12. Take $l = 7$ and $w = \frac{1}{2}$ in Proposition 13.3. Compute $M(S_{\mathrm{a}})$, $M(S_{\mathrm{b}})$ and their difference in terms of Q, and compare the difference with the blur $(k-1)Q$ required in Proposition 5.3 (c).

Exercise 13 (Insulating the label 0)**.** The introduction of IUT III insulates the "core label 0" from the label indeterminacies. Suppose that the permutations of S_{a} must fix 0, so that the theta value of level j may sit only in the factors $1, \ldots, j$. Show that for two places $M = -Q\, A_0(w)$ with $A_0(w) = \frac{1}{l^*} \sum_j j^2 w^{j-1}$. For $l = 7$ and $w = \frac{1}{2}$, does Cor hold for every $E \geq 0$? How large must E be?

Exercise 14. Explain why the two statements of Corollary 14.4 cannot be combined, by Theorem 14.3 (ii), into a bound on Q.

Exercise 15. Show that for a fixed $w < 1$ one has $A(w) \leq w(1+w)/\big(l^*(1-w)^3\big)$, so that $A(w) \to 0$ as $l \to \infty$. What does this say about $\mathrm{Cor}(S_{\mathrm{a}})$ for large l?

Answers

1. $x_B = 2x_A$ and $x_C = 6x_A$, and $x_A = \frac{1}{6}x_C$ holds for every x_A; the consistent measurements are exactly $(t, 2t, 6t)$, $t \in \mathbb{R}$. With $\frac{1}{5}$ the monodromy around the loop is $6/5 \neq 1$, so by Proposition 4.2 (ii) the only value is 0.

2. Reflexivity comes from id, symmetry from the inverse in (R1) and transitivity from the composite in (R2). There are three classes: the negative numbers, 0 and the positive numbers.

3. $k = (l^*+1)(2l^*+1)/6$ with $l^*+1 = (l+1)/2$ and $2l^*+1 = l$, so $k = l(l+1)/12$, and $k - 1 = (l^2 + l - 12)/12 = (l+4)(l-3)/12$. For $l = 5, 7, 11, 13$: $k = 5/2,\ 14/3,\ 11,\ 91/6$.

4. The condition on the edge $n \to n+1$ is $x_{n+1} = \lambda x_n$. Induction gives $x_n = \lambda^n x_0$ for $n \geq 0$, and induction using $x_n = \lambda^{-1} x_{n+1}$ gives it for $n < 0$. Conversely this family satisfies every condition.

5. For $n = -2, -1, 0, 1, 2$: $-4,\ -2,\ -1,\ -\frac{1}{2},\ -\frac{1}{4}$. As $n \to +\infty$ the elements approach 0; as $n \to -\infty$ they become arbitrarily small.

6. $A = \epsilon$. With the factor k: $(k-1)A = \epsilon$. With the inequality: $(k-1)A \le E$ (Proposition 6.1).

7. $k = 14/3$ and $k - 1 = 11/3$, so $Q \le 3/11$. For $k = 1$ the inequality reads $0 \cdot Q \le E$ and gives no bound. In the actual IV the constant C also depends on l, so the model says nothing about the behaviour as l grows.

8. The sign and the ratio (and the order relation) are unchanged. The absolute value and the difference are multiplied by λ_f (Proposition 3.7).

9. A product of scales equal to 1 is 1. As in the proof of Proposition 4.2 (iii), put values along a spanning tree; the conditions on the edges outside the tree then hold as well.

10. The sum over t is a product of $j+1$ sums over the places, one for each factor: $\sum_t w(t)\, Q_{t_i} = (\sum_v w_v Q_v) \prod_{i' \ne i} (\sum_v w_v) = Q$. Hence the average of b_j is $-j^2 Q$, and $\bar{\mu}(b) = -\frac{1}{l^*} \sum_j j^2 Q = -kQ$.

11. $A(w) = (w + 4w^2)/2$, and $A(w) \le 1$ means $4w^2 + w - 2 \le 0$, that is $w \le (\sqrt{33} - 1)/8 \approx 0.593$.

12. $k = 14/3$ and $A(\frac{1}{2}) = \frac{1}{3}(\frac{1}{2} + 1 + \frac{9}{8}) = \frac{7}{8}$, so $M(S_\mathrm{a}) = -\frac{7}{8}Q$, $M(S_\mathrm{b}) = -\frac{14}{3}Q$ and the difference is $\frac{91}{24}Q$. The required blur is $(k-1)Q = \frac{11}{3}Q = \frac{88}{24}Q$; the difference exceeds it by $\frac{1}{8}Q = (1 - A(\frac{1}{2}))\, Q$, so $\mathrm{Cor}(S_\mathrm{a})$ holds with $E = 0$. The ratio of the two is $91/88$.

13. The smallest q-order among the factors $1, \dots, j$ is Q_b if these j factors sit at the bad place, with weight w^j, so $M = -\frac{1}{l^*} \sum_j j^2 w^j Q_\mathrm{b} = -Q\, A_0(w)$. For $l = 7$ and $w = \frac{1}{2}$, $A_0 = \frac{1}{3}(1 + 2 + \frac{9}{4}) = \frac{7}{4} > 1$, so Cor does not hold for $E = 0$; it needs $E \ge (A_0 - 1)Q = \frac{3}{4}Q$. The difference with S_b is $(k - A_0)Q = \frac{35}{12}Q$, which is $35/44$ of the required $\frac{11}{3}Q$.

14. $\mathrm{Cor}(S_\mathrm{a})$ is a statement about $M(S_\mathrm{a})$ and $\mathrm{IV}(S_\mathrm{b})$ one about $M(S_\mathrm{b})$, whereas Theorem 14.3 (ii) needs one number in both. Chaining them would need $M(S_\mathrm{a}) \le M(S_\mathrm{b})$, the opposite of (i); in the model $M(S_\mathrm{a}) - M(S_\mathrm{b}) = (k - A(w))\, Q > 0$.

15. $\sum_{j \ge 1} j^2 w^j = w(1+w)/(1-w)^3$, so $A(w) \le w(1+w)/(l^*(1-w)^3) \to 0$. For large l, $A(w) \le 1$ for every fixed $w < 1$: at every prime with a good place, $\mathrm{Cor}(S_\mathrm{a})$ holds whatever Q is.

A Verification in Lean

The file `lean/Components.lean` checks the quantitative core of Part II in Lean 4 (version 4.30.0) with Mathlib. It treats one level j with an arbitrary finite set of places for Propositions 13.1 and 13.2, and two places, encoded as `Bool`, from Proposition 13.3 on. The statements $\mathrm{Cor}(S)$ and $\mathrm{IV}(S)$ are `Cor312` and `IVbound`. Every declaration depends at most on the axioms `propext`, `Classical.choice` and `Quot.sound`; there is no `sorry`. Proposition 1.7 of IUT IV, of which Proposition 13.1 (iv) is the case in the model, is formalized in [LANA-iut] (theorem `weighted_average_eq` in `Iut4Sec1/Combinatorics/WeightedAverage.lean`).

In the text	Lean declarations
Setup 12.1 (e), Exercise 3	`kmean_eq_1`, `kmean_gt_one`
Prop. 13.1 (i)	`hull_univ_theta`, `hull_univ_theta_min`
Prop. 13.1 (ii)	`hull_fix_theta`
Prop. 13.1 (iii)	`thetaSym_le_hull`, `symm_le_hull`
Prop. 13.1 (iv)	`tupleWeight_comp`, `wavg_act`, `wavg_factor`, `wavg_thetaSym`, `wavg_symm`
Prop. 13.2	`wavg_hull_univ`, `le_hull`, `hull_univ_const`; the converse (the difference is 0 only if all q-orders agree) is proved on paper only
Prop. 13.3	`hull_univ_two`, `wavg_hull_two`, `wavg_theta_two`, `muA_eq`, `muB_eq`
Lemma 14.2	`sum_sq_half`, `Afun_half_le_one`, `Afun_mono`, `Afun_le_one_of_le_half`
Thm. 14.3 (i)	`hull_mono`, `wavg_mono` (the general case is these two lemmas averaged over the levels); `muB_le_muA`, `cor312_A_of_B`, `iv_B_of_A` (two places)
Thm. 14.3 (ii)	`conservation`
Thm. 14.3 (iii)	`cor312_B_iff`, `iv_B_iff`
Thm. 14.3 (iv)	`cor312_A_of`, `cor312_A_iff`, `Afun_lt_kmean`, `iv_A_fails`
Thm. 14.3 (v)	`cor312_A_scaled`, `cor312_B_fails_scaled`
Cor. 14.4	`mixed_pair_holds`

The log in `logs/` records the output of `lake env lean Components.lean`, the `#print axioms` lines and the exit code.

Acknowledgements and use of AI

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